QualityForge 全项目产品需求文档

AI PRD

文档类型	版本	状态	日期
PRD	v1.1		2026-08-10

文档定位
QualityForge AI PRD OpenAPI DB AI

0.

	QualityForge AI
	/
	AI
	ISO/IEC/IEEE 29148 PRD Given/When/Then
	OpenAPI
	10 58

0.1

- 1
- 2
- 3
- 4
- 5
- 6
- 7 AI Agent Skill
- 8
- 9

- 10
- 11
- 12
-

1.

1.1

QualityForge AI AI CI/CD
AI

1.2

-
-
- Locator
- Trace
- CI
- AI oracle

1.3

	RequirementDocument → TestingIR → BusinessRule → Scenario → DesignTestCase → Assertion → Coverage
	AI Secret fail-closed
	Agent
	API DB

2.

“ ” Runner PRD

		AI
/		/ DesignTestCase ReviewTask Coverage
	Runner	Secret
		AI
/	Runner Secret	
AI/Skill	Agent Prompt Skill	

3.

3.1

Skill Agent CI/CD	CI
API Web UI	
Jira/Git/CI/	
AI	

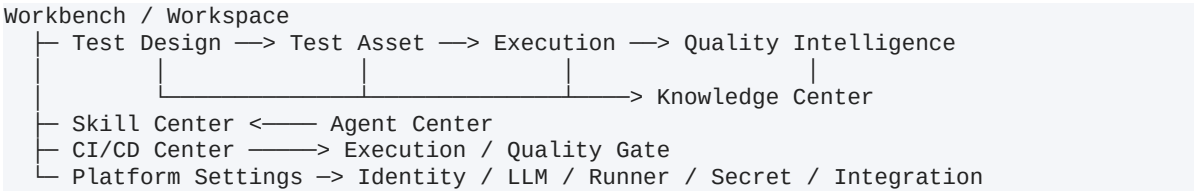
3.2

		80%	
	/P95	80%	oracle
	Top-K	95%	Recall/Precision
CI	TIA	60%	
	sourceTrace		Gate 0 100%
AI	AI		Skill/ /

□□□□

oracle 80%/95%/60%

4.



Workbench	AI
Workspace	Secret OpenAPI
Test Design	IR/
Test Asset	Locator Mock
Execution	Runner /
Quality	Coverage
AI	Skill Agent Prompt Workflow Trace
Knowledge / CI / Settings	

5.

5.1

- Workspace
- RequirementDocument
- REJECT PASS_WITH_REVIEW ReviewTask PASS
- TestingIR BusinessRule SourceTrace
- TestDesignPlan TestScenario DesignTestCase Assertion
- Coverage TestCase

5.2

- TestCase TestSuite ExecutionPlan Runner
- REST WebSocket
- ExecutionResult Trace
- Quality Intelligence Coverage RCA

- Knowledge Center CI/CD QualityGate

5.3 Agent/Skill Gate 0

```
Supervisor → Requirement Quality Gate → / /
→ Requirement Compiler → Independent IR Validator → / / /
→ Assertion Verifier → Coverage Analyzer → HITL Gatekeeper →
```

6.

6.1 Workspace

AI

V1		Workspace WorkspaceEnvironment WorkspaceSecret WorkspaceAIContext	workspaceId/tenantId

6.1.1

- CI
- Dev/Test/Staging URL
- AI
-

6.1.2

- Workspace Workspace
- Secret SecretRef
- AI
-

6.1.3

1

目标：

主要功能

-
-
-
- Active/Archived

设计理由：

2

目标：

主要功能

-
- /API OpenAPI
-
- Test Design Skill QI Prompt

设计理由： AI

3

目标：

主要功能

- Dev/Test/Staging Base URL
-
- SecretRef
- Workspace Skill

设计理由： Secret

4 AI

目标： AI

主要功能

- URL OpenAPI/Swagger
- Schema
- WorkspaceAIContext sourceTrace
-

设计理由： AI

6.1.4

	OpenAPI
	Workspace SecretRef WorkspaceAIContext
	Platform Settings Secret LLM
	Test Design Test Asset Execution Knowledge CI/CD

6.1.5

ID			
QF-FR-WS-001	P0		
QF-FR-WS-002	P0		
QF-FR-WS-003	P0		Dev/Test/Staging Secret
QF-FR-WS-004	P1	AI	OpenAPI/Swagger

6.2 Test Design

Testing IR

V1		RequirementDocument TestingIR BusinessRule TestDesignPlan TestScenario DesignTestCase Assertion ReviewTask	TestCase REJECT/BLOCK_HUMAN

6.2.1

-
-
-
- AI
- “ ”

6.2.2

- → Gate → Testing IR → / → → → → Coverage
- sourceTrace Gate PASS
- DesignTestCase TestCase
- Agent ReviewTask
- Coverage AI

6.2.3

1

□□□

□□□□

- PDF Word Markdown OpenAPI
-
- AI
-

□□□□□

2

目标： Testing IR

主要功能

- NFR Finding
- Finding
- NFR AI
- PASS PASS_WITH_REVIEW REJECT ReviewTask

设计理由： Gate

3 Testing IR

目标：

主要功能

- NFR
- BusinessRule ruleId
- IR Validator Schema sourceTrace
- IR

设计理由： IR

4

目标：

主要功能

- UI
- /
- Workspace
- RequirementDocument/TestingIR

设计理由：

5

目标：

主要功能

-
-
-

-

设计理由：

6

目标：

主要功能

- DesignTestCase sourceTrace
- Assertion Schema UI
-
-
- TestCase

设计理由： Coverage

6.2.4

	RequirementDocument OpenAPI WorkspaceAIContext Knowledge Pattern
	TestingIR BusinessRule TestDesignPlan TestScenario DesignTestCase Assertion Coverage ReviewTask
	Workspace Knowledge AI Agent/Skill Gate 0
	Test Asset Execution Quality Intelligence

6.2.5

ID			
QF-FR-TD-001	P0		PDF Word Markdown OpenAPI
QF-FR-TD-002	P0		NFR
QF-FR-TD-003	P0	Testing IR	IR Source Trace
QF-FR-TD-004	P0		
QF-FR-TD-005	P0		DesignTestCase Assertion TestCase
QF-FR-TD-006	P1		

6.3 Test Asset

Mock

V1		TestCase TestSuite Assertion Locator TestDataProfile APIContract MockConfig	AI

6.3.1

- Mock
-
- UI
-
- Mock

6.3.2

- TestCase
-
- AI
- Mock Schema

6.3.3

1

目标： TestCase

主要功能

-
-
- AI
-

设计理由：

2

目标：

主要功能

- /

-
-
-

设计理由：

3

目标：“ ”

主要功能

-
- API Schema UI
- VERIFIED/REJECTED/UNVERIFIED
-

设计理由：

4 Locator

目标： UI

主要功能

-
- DOM Locator
-
- before/after

设计理由： Locator “ ”

5

目标：

主要功能

- CSV/JSON
-
- Secret
-

设计理由：

6 API Mock

□□□ Schema

□□□□

- OpenAPI/Swagger Schema
-
- Schema Mock
- Mock

□□□□□ Mock

6.3.4

	DesignTestCase Assertion API Contract Locator
	TestCase/TestSuite Mock
	Test Design Workspace Platform Settings
	Execution Quality Intelligence CI/CD

6.3.5

ID			
QF-FR-TA-001	P0		TestCase CRUD CSV
QF-FR-TA-002	P0		
QF-FR-TA-003	P1	Locator	
QF-FR-TA-004	P1		
QF-FR-TA-005	P1	Mock	API Contract /

6.4 Execution

API/UI Trace

V1	Runner	ExecutionPlan ExecutionTask ExecutionResult RunnerNode	HTTP

6.4.1

- Runner
- API/UI/
- Trace
- Runner
- HTTP 0

6.4.2

- Runner/Skill
- Skill Runner
- attempt
- RCA QI

6.4.3

1

目标：

主要功能

- TestSuite/TestCase Runner
- Cron Webhook CI/CD POST
- Secret Runner
- ExecutionPlan

设计理由：

2

目标：

主要功能

- Pending/Queued/Running/Completed/Failed/Cancelled

-
-
-

设计理由： Runner CI

3 Runner

目标：

主要功能

- Runner Token OS Skill
- OnlineIdle/OnlineBusy/Offline/HeartbeatLost
-
-

设计理由：

4 Skill

目标：

主要功能

- Skill Schema allowlist
- HTTP Playwright
- Secret Tool
- HTTP/

设计理由： Skill

5

目标：

主要功能

- WebSocket CaseResult
- stdout/stderr HTTP TraceId
- seq REST
- / RCA

设计理由： QI

6

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□□□□

- / /
-
- Runner
- attempt

□□□□□ fail-closed attempt

6.4.4

	ExecutionPlan TestCase/Suite Skill SecretRef
	ExecutionTask ExecutionResult /Trace/
	Workspace Test Asset Skill Center Platform Settings CI/CD
	Quality Intelligence CI/CD Knowledge

6.4.5

ID			
QF-FR-EX-001	P0		CI
QF-FR-EX-002	P0	Runner	Runner
QF-FR-EX-003	P0		API/UI
QF-FR-EX-004	P0		WebSocket
QF-FR-EX-005	P1		

6.5 Quality Intelligence

V1 / V2		QualityAnalysisReport FailureAnalysisDetail FailureCluster CoverageReport	AI

6.5.1

- Pass/Fail Trace
-
- Flaky Test
- AI oracle
-

6.5.2

- QI Execution
- Trace
- AI RCA
- Coverage

6.5.3

1

目标：

主要功能

-
- ExecutionResult
-
-

设计理由：

2

目标： Runner Skill

主要功能

- HTTP

- Secret Token
- Runner
- FailureAnalysisDetail RCA

设计理由： AI Prompt

3

目标：

主要功能

- ProductBug ScriptIssue EnvIssue DataIssue DependencyIssue
-
-
- manualVerified

设计理由： AI / /

4

目标：

主要功能

- Trace
-
-
- Jira

设计理由： Bug

5 **Flaky**

目标：

主要功能

-
- Flaky / /
-
- MTTR

设计理由： Flaky

6 Coverage

□□□

□□□□

- CoverageReport gate verdict
-
-
- CI/CD AI

□□□□□

6.5.4

	ExecutionResult /Trace/ Coverage Git Knowledge
	QualityAnalysisReport FailureAnalysisDetail FailureCluster CoverageReport RCA
	Execution Workspace Knowledge
	CI/CD Test Design Knowledge

6.5.5

ID			
QF-FR-QI-001	P0		
QF-FR-QI-002	P0		Trace
QF-FR-QI-003	P0		CoverageReport
QF-FR-QI-004	P1	RCA	
QF-FR-QI-005	P2		

6.6 Skill Center

Playwright HTTP Skill

V1 / V2	Skill	Skill SkillInvocation	/ / PASS allowlist Skill

6.6.1

- API UI Agent
-
- “Skill ” “ ”
- Skill

6.6.2

- Skill Agent Skill
- Manifest Must Not Schema allowlist
- Skill Workspace
- Skill eval

6.6.3

1

目标：

主要功能

- Skill
- Workspace / Skill
- eval
- Agent/Workflow

设计理由： “ ”

2 Manifest

目标： Skill

主要功能

- Manifest ID Schema
- SemVer

-
- Secret

设计理由： Manifest Agent Runner

3

目标： Skill

主要功能

- Schema
- evals/Golden fixtures
-
- Active

设计理由： checker PASS

4

目标： Agent/Execution Skill

主要功能

- Workspace Runner Skill
- Schema
- SecretRef
-

设计理由： Agent

5

目标：

主要功能

- CPU
-
- Secret
-

设计理由： Prompt

6

□□□

□□□□

- /Agent Workspace Runner ErrorCode
-
- P50/P95
- Secret

□□□□□ Skill Runner

6.6.4

	Manifest Skill Workspace Runner
	Skill SkillInvocation
	Platform Settings Workspace Agent Center Execution
	Agent Workflow Execution Quality Intelligence

6.6.5

ID			
QF-FR-SK-001	P0	Skill	API Testing Web Playwright
QF-FR-SK-002	P0		Skill Must Not
QF-FR-SK-003	P1		Skill eval
QF-FR-SK-004	P1		Secret CPU

6.7 Agent Center Agent

Agent Prompt Trace

Gate 0 / V2	AI	Agent AgentPromptVersion AgentSession AgentTraceNode	Agent

6.7.1

- AI Prompt
- Agent Gate
- “AI ”
-

6.7.2

- Gate 0 1 Supervisor + 3 Subagent + DAG Agent
- Agent allowlist Skill Prompt
- AI
- Trace

6.7.3

1 Agent Prompt

目标： Agent

主要功能

- Agent ID Skill
- Prompt
- Workflow
- V0 qf-supervisor qf-requirement-ir qf-test-design qf-independent-verifier

设计理由： Prompt Agent “ Agent”

2 Workflow

目标： DAG Agent Skill

主要功能

-
- Supervisor 1

- PASS REJECT/ERROR/TIMEOUT BLOCK_HUMAN
- V0 JSON DAG

设计理由： DAG

3

目标： Agent

主要功能

- runId/sessionId
- Skill Workflow Prompt /
-
- attempt

设计理由： Skill

4 Tool

目标： Agent Skill

主要功能

- allowedSkills Schema Secret
-
- Skill/ 1
-

设计理由： Agent

5 Trace

目标：

主要功能

- Agent Skill Finding Code
- CoT
- Tool
- runId

设计理由： “ ” Trace

6

□□□ Agent

□□□□

- /
- Agent/Prompt/ /Skill Golden Set
- Trace
- oracle

□□□□□ Token/

6.7.4

	Agent/Workflow Skill
	AgentSession AgentTraceNode
	Platform Settings Knowledge Skill Center AI
	Test Design Execution Quality Intelligence

6.7.5

ID			
QF-FR-AG-001	P0	Agent	Prompt Skill allowlist
QF-FR-AG-002	P0	Trace	
QF-FR-AG-003	P0	Gate 0	1 Supervisor + 3 Subagent + 13 Skill DAG
QF-FR-AG-004	P1		DAG

6.8 Knowledge Center

RAG /UI/ Agent

V1		KnowledgeItem KnowledgeVector BaselineItem	

6.8.1

-
- AI
- Schema
- AI
- RAG

6.8.2

- “ ”
- + + Workspace
- AI Draft/PendingReview Published
-

6.8.3

1

目标：

主要功能

- PDF Word Markdown OpenAPI
-
- Workspace
-

设计理由：

2

目标：

主要功能

- Schema

- Embedding
- Workspace
- Top-K

设计理由：

3 Testing Pattern

目标：

主要功能

-
- RCA Bug TestCase /
-
- Test Design QI ID

设计理由：

4

目标： API

主要功能

- API Response Schema UI /
- Diff
-
-

设计理由： “ ”

5 AI

目标：

主要功能

- RCA Pattern
-
-
- Agent RAG

设计理由： AI

6

目标：

主要功能

-
-
- Workspace
-

设计理由： RAG

6.8.4

	OpenAPI RCA/Bug
	KnowledgeItem KnowledgeVector BaselineItem
	Workspace QI Test Asset
	Test Design Agent QI Execution

6.8.5

ID			
QF-FR-KN-001	P0		
QF-FR-KN-002	P0		AI
QF-FR-KN-003	P0		API Schema
QF-FR-KN-004	P1		Testing Pattern

6.9 CI/CD Center

TIA

V1 / V2		PipelineConfig PipelineRun QualityGate	AI

6.9.1

-
- Flaky
- RCA PR/MR
- Webhook
- AI

6.9.2

- QualityForge CI Jenkins/GitLab/GitHub
- TIA
- QualityGate AI
- Webhook

6.9.3

1 Webhook

目标：

主要功能

- Jenkins/GitLab/GitHub
- timestamp nonce Secret
- provider/eventId/commit
- Workspace PipelineConfig ExecutionPlan

设计理由： CI

2

目标：

主要功能

- PipelineRun commit ExecutionTask

-
-
-

设计理由： CI “ ”

3 TIA

目标：

主要功能

- Git Diff
- BusinessRule APIContract TestCase
- / /
- /

设计理由：

4

目标：

主要功能

- / Coverage
-
- Passed/Warned/Blocked/Quarantined
-

设计理由：

5 Flaky

目标：

主要功能

- Flaky Quarantine
-
-
- Flaky

设计理由：

6

目标：

主要功能

- PR/MR Coverage
- RCA AI
-
- Scope

设计理由： PR

6.9.4

	Webhook/Git Diff Execution QI QualityGate
	PipelineRun PR/
	Workspace Platform Settings Execution QI
	CI Git PR/MR

6.9.5

ID			
QF-FR-CI-001	P0	Webhook	Jenkins/GitLab/GitHub
QF-FR-CI-002	P0		Passed/Warned/Blocked
QF-FR-CI-003	P1	TIA	Git Diff
QF-FR-CI-004	P1		

6.10 Platform Settings

LLM Runner Secret

V1	Runner	User Role LlmConfig RunnerNode WorkspaceSecret IntegrationConfig	

6.10.1

- LLM Runner Secret
-
- Runner
- Prompt
-

6.10.2

- Platform Settings Runner Secret
- tenant/workspace
- LLM Runner
- Secret

6.10.3

1

目标：

主要功能

-
- RBAC
-
- Workspace

设计理由：

2 LLM

目标： Agent AI

主要功能

- TPM API Key

-
- Token /Prompt
-

设计理由：

3 Runner

目标：

主要功能

- /Token Runner OS Skill
- CPU/
- Runner Token Token
- Execution

设计理由： Execution

4 Secret

目标： LLM Git CI

主要功能

-
- KEK/DEK
-
- SecretRef

设计理由： SecretRef

5

目标： Jira Git CI LDAP/SSO

主要功能

- provider OAuth/API Key/SSH SecretRef Scope
- Webhook
- Workspace / /
-

设计理由：

6

目标：

主要功能

- Secret Runner Skill
- /Workspace
-
-

设计理由：

6.10.4

	/ Runner Secret
	Runner SecretRef IntegrationConfig
	KMS/Vault Runner
	Workspace Agent Skill Execution Knowledge CI/CD

6.10.5

ID			
QF-FR-PS-001	P0		RBAC
QF-FR-PS-002	P0	LLM	
QF-FR-PS-003	P0	Runner	
QF-FR-PS-004	P0	Secret	

7. AI Agent Skill

7.1

L0	AI /	
L1	AI	
L2		allowlist
L3		Gate 0

7.2 Agent

Agent		Skill	
qf-supervisor		plan-orchestrator	
qf-requirement-ir	IR		IR
qf-test-design			IR
qf-independent-verifier		IR Coverage HITL	

7.3

- docs Skill Golden
- deny_all Secret
- Skill Schema allowlist
- 1 Schema fail-closed
- Prompt
-

8.

8.1

- tenant/workspace
- / SourceTrace
- DesignTestCase TestCase

- CoverageReport
- /Prompt/Skill

8.2

REST	/api/v1 camelCase JSON Bearer/API Key/Runner Token
WebSocket	Execution AI Agent Trace Workspace seq
CI/CD	Webhook
	URL
LLM	

9.

	V1	/
	50+ API FCP < 3.0s AI P95 < 20s	Runner
	AES-256-GCM RBAC	RLS/ Secret
	1 Runner	V2
	JSON Metrics	requestId/runId/traceId Agent
		Saga
		WCAG 2.2 AA +
		Prompt

10.

Gate 0 / V0		→IR→ → → → →Coverage→	
V1 1–3		Workspace / API/UI Skill/Agent RAG/Webhook	Runner Skill Agent
V2 4–6		Locator RCA Skill Workflow TIA/	
V3 7–12		Runner Skill Agent PR	

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B-1/B-2/B-3	2026-07-31	2026-08-04	B-4/B-5/B-6	Sprint 0
OpenAPI operationId AI /				

11.

11.1

AC-E2E-01

Given **When** Gate IR **Then** Gate PASS TestCase

AC-SEC-01

Given A B **When** A Token B ID **Then** 403 404

AC-AI-01

Given Agent IR/ / **When** Agent **Then**

AC-EX-01

Given Runner **When** Runner **Then** HTTP 2xx PASS

AC-CI-01

Given **When** **Then**

11.2

- ID /API/ / 2xx
- OpenAPI operationId / / / / /
- PRD API DB
- AI oracle Golden Set Prompt Skill
- NFR
- /

12.

ID			
DEC-01			RBAC /API/DB
DEC-02	164 OpenAPI operationId	SDK	Sprint 0 operationId
DEC-03	AI	PASS/REVIEW	Golden Set
DEC-04	LLM oracle		checker + +

ID			
DEC-05		ROI	
DEC-06			Gate 0/V1 V2/V3
DEC-07			

A.

	54- .md	
	03- Glossary.md	AI
	04- .md	
	10–19 PRD	
AI	21-AI .md	L0–L3
Agent/Skill	55-Agent Skill SOP.md	allowlist
API	32-API .yaml	REST Schema
DB	33- Schema .md	
/	34- .md / 36-WebSocket .md	RBAC/RLS/Secret
	docs/05-audits/current/	

03/04 PRD 04 32 33 AI 21 03 04 32 33 PRD